

Problem 1.

Solution: We generate $n = 1000$ random samples $U_i \sim \text{Unif}(0, 1)$, and define

$$Y_i^{(1)} = e^{U_i}, \quad Y_i^{(2)} = \sqrt{1 - U_i^2}, \quad i = 1, 2, \dots, n.$$

The sample means are

$$\bar{U} = \frac{1}{n} \sum_{i=1}^n U_i, \quad \bar{Y}^{(1)} = \frac{1}{n} \sum_{i=1}^n Y_i^{(1)}, \quad \bar{Y}^{(2)} = \frac{1}{n} \sum_{i=1}^n Y_i^{(2)}.$$

Then the estimated covariances and variances are computed as:

$$\widehat{\text{Cov}}(U, e^U) = \frac{1}{n} \sum_{i=1}^n (U_i - \bar{U})(Y_i^{(1)} - \bar{Y}^{(1)}) = \frac{1}{n} \sum_{i=1}^n (U_i Y_i^{(1)}) - \bar{U} \bar{Y}^{(1)},$$

$$\widehat{\text{Cov}}(U, \sqrt{1 - U^2}) = \frac{1}{n} \sum_{i=1}^n (U_i - \bar{U})(Y_i^{(2)} - \bar{Y}^{(2)}) = \frac{1}{n} \sum_{i=1}^n (U_i Y_i^{(2)}) - \bar{U} \bar{Y}^{(2)}.$$

$$\widehat{\text{Var}}(U) = \frac{1}{n} \sum_{i=1}^n (U_i - \bar{U})^2, \quad \widehat{\text{Var}}(Y^{(2)}) = \frac{1}{n} \sum_{i=1}^n (Y_i^{(2)} - \bar{Y}^{(2)})^2.$$

The sample correlation estimate is therefore

$$\widehat{\text{Corr}}(U, \sqrt{1 - U^2}) = \frac{\widehat{\text{Cov}}(U, \sqrt{1 - U^2})}{\sqrt{\widehat{\text{Var}}(U)} \sqrt{\widehat{\text{Var}}(Y^{(2)})}.$$

Using Python with $n = 1000$ runs:

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1 Estimated Cov(U, e^U): 0.1426
2 Estimated Corr(U, sqrt(1 - U^2)): -0.9206
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Problem 2.

Solution:

(a) Set $X = 0$ (initial state). For $n = 1, 2, \dots, N - 1$:

If $X = 0$: generate next X from $\{0, 1, 2\}$ with probabilities $(\frac{1}{2}, \frac{1}{3}, \frac{1}{6})$,

If $X = 1$: generate next X from $\{0, 1, 2\}$ with probabilities $(0, \frac{1}{3}, \frac{2}{3})$,

If $X = 2$: generate next X from $\{0, 1, 2\}$ with probabilities $(\frac{1}{2}, 0, \frac{1}{2})$.

(b) Set $X = 0$ (initial state). For $n = 1, 2, \dots, N - 1$:

If $X = 0$: generate next X from $\{0, 1, 2\}$ with probabilities $(\frac{1}{2}, \frac{1}{3}, \frac{1}{6})$,

If $X = 1$: generate next X from $\{0, 1, 2\}$ with probabilities $(0, \frac{1}{3}, \frac{2}{3})$,

If $X = 2$: generate next X from $\{0, 1, 2\}$ with probabilities $(\frac{1}{2}, 0, \frac{1}{2})$.

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(c) Estimator:

$$\hat{P}(X_{10} = 1 \mid X_0 = 0) = \frac{1}{M} \sum_{m=1}^M \mathbf{1}\{X_{10}^{(m)} = 1\}$$

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1  Set M = 100
2  For m = 1, 2, ..., M:
3      Set X = 0
4      For t = 1, 2, ..., 10:
5          Given current state X = i,
6              generate next state X = j with probability Pij
7          Set X = j
8      End For
9      Record Im = I{X = 1}
10 Compute P_hat = (1/M) * sum(Im)

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Result:

$$\hat{p} \approx 0.21$$

(d) Estimator:

$$\hat{E}[X_{10} \mid X_0 = 0] = \frac{1}{M} \sum_{m=1}^M Y_m$$

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1  Set M = 100
2  For m = 1, 2, ..., M:
3      Set X = 0
4      For t = 1, 2, ..., 10:
5          Update X to j with probability Pij
6      End For
7      Record Ym = X
8  Compute E_hat = (1/M) * sum(Ym)

```

Result:

$$\hat{E}[X_{10} \mid X_0 = 0] \approx 1.01$$

(e) Estimator:

$$\hat{P}(T \leq 10 \mid X_0 = 0) = \frac{1}{M} \sum_{m=1}^M \mathbf{1}\{T^{(m)} \leq 10\}$$

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1  Set M = 100
2
3  For m = 1, 2, ..., M:
4      Set X = 0, t = 0
5      While t < 10 and X != 1:
6          Generate next state X = j with probability Pij
7          Set t = t + 1
8      End While
9      Record Im = Indicator(X == 1)
10 End For
11
12 Compute P_hat = (1/M) * sum(Im)

```

Result:

$$\hat{P}(T \leq 10 \mid X_0 = 0) \approx 0.75$$

(f) Estimator:

$$\hat{E}[T \mid X_0 = 0] = \frac{1}{M} \sum_{m=1}^M T^{(m)}$$

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1  Set M = 100
2
3  For m = 1, 2, ..., M:
4      Set X = 0, t = 0
5      While X != 1:
6          Generate next state X = j with probability Pij
7          Set t = t + 1
8      End While
9      Record Tm = t
10 End For
11
12 Compute E_hat = (1/M) * sum(Tm)

```

Result:

$$\hat{E}[T \mid X_0 = 0] \approx 3.43$$

(g) Long-run state proportions:

$$\hat{\pi}_i = \frac{1}{N-b} \sum_{t=b+1}^N \mathbf{1}\{X_t = i\}, \quad i = 0, 1, 2$$

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1  Set simulation length N = 10000
2  Set burn-in b = 100
3  Set X = 0
4
5  For t = 1, 2, ..., N:
6      Generate next state X = j with probability Pij
7      Set X = j
8      Record state X if t > b
9  End For
10
11 Compute pi_hat(i) = (1 / (N - b)) * sum(Indicator(Xt == i)) for i
    = 0, 1, 2

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Result:

$$\hat{\pi}_0 = 0.4068, \quad \hat{\pi}_1 = 0.1948, \quad \hat{\pi}_2 = 0.3984$$

(h) Long-run average cost:

$$\hat{g} = \frac{1}{N-b} \sum_{t=b+1}^N c(X_t), \quad \text{where } c(0) = 1, c(1) = 3, c(2) = 2$$

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1  Use the same trajectory as (g)
2
3  Set costs: c(0)=1, c(1)=3, c(2)=2
4  Compute g_hat = (1 / (N - b)) * sum(c(Xt) for t = b+1,...,N)

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Result:

$$\hat{g} \approx 1.7881$$

Problem 3.

Solution:

- (a) Given $X_n = x$, among the x active individuals: the number who remain active in the next period follows

$$A_{n+1} \sim \text{Binomial}(x, \alpha).$$

Among the $N - x$ inactive individuals: the number who become active in the next period follows

$$B_{n+1} \sim \text{Binomial}(N - x, 1 - \beta).$$

By independence of individual transitions,

$$X_{n+1} = A_{n+1} + B_{n+1}.$$

Hence, the conditional distribution of X_{n+1} depends only on $X_n = x$, not on any earlier history. Therefore, $\{X_n, n \geq 0\}$ is a time-homogeneous discrete-time Markov chain on the state space $\{0, 1, 2, \dots, N\}$.

For $y = 0, 1, 2, \dots, N$, the transition probabilities are:

$$P(X_{n+1} = y \mid X_n = x) = \sum_{k=\max(0, y-(N-x))}^{\min(x, y)} \binom{x}{k} \alpha^k (1-\alpha)^{x-k} \binom{N-x}{y-k} (1-\beta)^{y-k} \beta^{N-x-(y-k)}.$$

Thus, X_n is a binomially driven Markov chain describing the total number of active individuals in the population.

- (b) Estimator:

$$\hat{X} = \frac{1}{T-b} \sum_{t=b+1}^T X_t$$

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1  Set N = 100, alpha = 0.2, beta = 0.4
2  Set total time T = 10000, burn-in b = 100
3  Initialize X = 10, running_sum = 0
4
5  For t = 1, 2, ..., T:
6      # Active survivors
7      Generate Ui ~ Unif(0, 1) for i = 1, ..., X
8      Set S = sum(Indicator(Ui <= alpha))

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9
10      # Inactive -> active transitions
11      Generate Vj ~ Unif(0, 1) for j = 1, ..., (N - X)
12      Set A = sum(Indicator(Vj <= (1 - beta)))
13
14      # Update total number of active individuals
15      Set X = S + A
16
17      If t > b:
18          running_sum = running_sum + X
19      End For
20
21      Compute X_hat = running_sum / (T - b)

```

Result:

$$\hat{X} \approx 42.80$$

(c) Fix n and condition on $(Y_n, Z_n) = (y, z)$.

Initially active cohort (size 10): Among the y currently active individuals, the number that remain active in the next period follows

$$\text{Binomial}(y, \alpha),$$

and among the $10 - y$ currently inactive individuals, the number that become active follows

$$\text{Binomial}(10 - y, 1 - \beta).$$

Hence,

$$Y_{n+1} \mid Y_n = y \sim \text{Binomial}(y, \alpha) + \text{Binomial}(10 - y, 1 - \beta).$$

Initially inactive cohort (size $N - 10$): Among the z currently active individuals, the number that stay active follows

$$\text{Binomial}(z, \alpha),$$

and among the $N - 10 - z$ currently inactive individuals, the number that become active follows

$$\text{Binomial}(N - 10 - z, 1 - \beta).$$

Thus,

$$Z_{n+1} \mid Z_n = z \sim \text{Binomial}(z, \alpha) + \text{Binomial}(N - 10 - z, 1 - \beta).$$

Joint behavior: Because the initially active and inactive groups are disjoint and their members transition independently, Y_{n+1} and Z_{n+1} are conditionally independent given $(Y_n, Z_n) = (y, z)$. Therefore, for any (y', z') ,

$$P((Y_{n+1}, Z_{n+1}) = (y', z') \mid Y_n = y, Z_n = z) = P(Y_{n+1} = y' \mid Y_n = y) P(Z_{n+1} = z' \mid Z_n = z).$$

(d)

(e) Estimator:

$$\hat{p} = \frac{1}{T-b} \sum_{t=b+1}^T \mathbf{1}\{Y_t > Z_t\}$$

```

1  Set N = 100, alpha = 0.2, beta = 0.4
2  Set total time T = 10000, burn-in b = 1000
3  Set K = 10 # size of initially active cohort
4  Initialize Y = 10, Z = 0, Sum_Indicator = 0
5
6  For t = 1, 2, ..., T:
7
8      # Initially active cohort (size K = 10)
9      Generate Ui ~ Unif(0, 1) for i = 1, ..., Y
10     Set Sr = sum(Indicator(Ui <= alpha))
11
12     Generate Vj ~ Unif(0, 1) for j = 1, ..., (K - Y)
13     Set Ar = sum(Indicator(Vj <= 1 - beta))
14
15     Set Y = Sr + Ar
16
17     # Initially inactive cohort (size N - K)
18     Generate Ui ~ Unif(0, 1) for i = 1, ..., Z
19     Set Sz = sum(Indicator(Ui <= alpha))
20
21     Generate Vj ~ Unif(0, 1) for j = 1, ..., (N - K - Z)
22     Set Az = sum(Indicator(Vj <= 1 - beta))
23
24     Set Z = Sz + Az
25
26     # After burn-in, check if more active individuals
27     # come from the initially active cohort
28     If t > b:
29         Sum_Indicator = Sum_Indicator + Indicator(Y > Z)
30
31 End For
Compute p_hat = Sum_Indicator / (T - b)

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Result:

$$\hat{P}(Y_n > Z_n) \approx 0.0$$